NANYANG TECHNOLOGICAL UNIVERSITY
SINGAPORE

SC2006 – Software Engineering

Lab 1 Deliverables

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1.Functional Requirements

1. User Account Management

Description:

The system shall allow **Users** to create and manage their accounts securely.

Functional Requirements:

1.1 The system shall provide a registration form with mandatory fields for Email Address, Username, Password and car type and model.

- 1.1.1 The system shall permit usernames to be non-unique.
- 1.1.2 The system shall enforce a password policy requiring a minimum of 8 characters, including at least one uppercase letter, one lowercase letter, and one special character.
- 1.1.3 The system shall validate that the email address follows the format of address@domain.com

1.2 The system shall verify the **User's** email during registration before activation.

- 1.2.1 The system shall send a unique verification link to the **user's** email address, activating the account only upon successful click within 24 hours.

1.3 The system shall allow **Users** to log in using valid credentials.

- 1.3.1 The system shall provide a login form with mandatory fields for Email Address & Password.
- 1.3.2 The system shall authenticate users by matching provided credentials with the registered account database.
- 1.3.3 The system shall establish an authorized session for the user upon successful credential match.
- 1.3.4 The system shall display specific error messages for failed validation defined in 1.3.1 and 1.3.3
- 1.3.5 The system shall provide a “register account” button that redirects user to Register Page

1.4 The system shall allow users to log out at any time.

- 1.4.1 The system shall provide a "Logout" button on the navigation header that redirects user to the login page.
- 1.4.2 The system shall revoke the user’s authorised sessions upon logging out.

1.5 The system shall provide a “forget password” button that redirects users to the Reset Confirmation page.

   1.5.1 The system shall provide a reset confirmation form with mandatory fields for Email Address.

   1.5.2 The system shall validate that the email address follows the format of address@domain.com.

   1.5.3 Upon selection of the "Confirm" button, the system shall verify if the Email Address exists in the user database.

   1.5.4 The system shall send a Reset Password page URL to the provided Email Address if the address is successfully verified.

   1.5.5 The system shall display an error message if provided Email Address is not found in the database.

1.6 The system shall have a Reset Password page.

   1.6.1 The system shall provide a reset password form with mandatory fields for new password, confirmed new password.

   1.6.2 The system shall validate that the new password and confirmed new password are identical.

   1.6.3 Upon successful validation, the system shall update the user's account with the new password and invalidate the reset token.

   1.6.4 Upon successful validation, the system shall display a success message and redirect the user to the Login page.

   1.6.5 The system shall enforce a password policy requiring a minimum of 8 characters, including at least one uppercase letter, one lowercase letter, and one special character.

   1.6.6 The system shall display specific descriptive error messages for any failed validation defined in 1.7.1, 1.7.2 and 1.7.5.

1.7 The system shall allow **Users** to view their personal profile (e.g., username, car type and model and email address).

   1.7.1 The system shall provide a "Profile" button on the navigation header that redirects **Users** to Profile page.

   1.7.2 The system shall display user details, username, car type and model, email address.

   1.7.3 The system shall display allow users to change username, car type and model.

   1.7.4 Upon selection of "Save”, the system shall update the user's record in the database.

1.7.5 Upon selection of "Save”, the system shall display a "Profile Updated Successfully" message.

1.7.6 The system shall allow users to view their wallet balance in the profile page.

1.7.7 The system will allow users to logout defined in 1.4

1.7.8 The system will allow users to view company contact details (support email, phone number, address).

2. Location Search and Map Display

Description:

The system shall provide an interactive map that displays carparks and EV charging stations near a **User's** location or chosen destination.

Functional Requirements:

2.1 The system shall display a map upon application launch.

2.1.1 The system shall display carpark and EV charging station markers on the map depending on the user’s choice.

2.1.2 The system shall support zoom, pan and selection of markers on the map.

2.2 The system shall request permission to access the user’s location at app launch for the first time.

2.2.1 The system shall retrieve the user’s current location when permission is granted.

2.2.2 The system shall display a message for 3 seconds when location is not granted.

2.2.3 The system shall log user out when location location is not granted after the message is displayed.

2.2.4 The system shall continue to prompt user for permission to access the user’s location upon opening the app again.

2.3 The system shall provide a search bar with a search button that allows users to find destinations.

2.3.1 The system shall support partial matching when searching by location name or address.

2.3.2 The system shall support exact matching when searching by postal code.

2.3.3 The system shall display matching search results upon clicking search button.

2.3.4 The system shall display a message when no matching destinations are found.

2.3.5 The system shall allow users to select a destination from search results.

2.3.6 The system shall center the map around the selected destination.

2.4 The system shall display carparks or EV charging stations within a search area radius of 2 km around the current or selected destination.

3. Preference-Based Recommendation (Core Logic)

Description:Functional Requirements:

The system shall recommend optimal carparks or EV chargers based on user-defined preferences.

Functional Requirements:

3.1 The system shall allow the **User** to select one main preference:

- Cheapest
- Nearest to destination
- Most available

3.2 The system shall evaluate available options based on a weighted algorithm using the following parameters:

- Price
- Distance to destination
- Availability

3.3 The system shall display the top 3 recommended options with a “Reccomended” tag.

4. Carpark features

Description:

The system shall provide detailed, real-time carpark information and navigation support.

Functional Requirements:

4.1 The system shall allow the **user** to filter for carparks which displays carparks only.

4.2 The system shall display real-time carpark availability/vacancy using colour coded indicators. (Green = Available, Yellow = Limited-slots, Red = full)

4.3 The system shall allow **users** to click on a specific carpark for more details.

4.3.1 The system shall display details of the carpark.

- Name
- Carpark Vacancy
- Pricing

4.4 The system shall allow **Users** to find the nearest or cheapest available carpark.

4.4.1 The system shall retrieve carparks based on the user's location or selected location.

4.4.2 The system shall recommend carparks using the same preference-based recommendation logic (3. Preference-Based Recommendation)

4.4.3 The system shall display search results as a sorted list.

4.4.3.1 The system shall display details of the carpark as in 4.3.1.

4.4.4 The system shall display a message when no carparks are found.

5. EV charging features

Description:

The system shall enable EV-related search, recommendation, and booking functionalities.

Functional Requirements:

5.1 The system shall allow the **Users** to filter for EV charging stations which display EV charging stations only.

5.2 The system shall display real-time availability of EV charging stations.

5.3 The system shall allow **Users** to click on a specific EV charging station for more details.

5.3.1 The system shall display details of the EV charging station.

- Station Name
- Adapter type
- EV charging provider
- Price

5.4 The system shall allow **Users** to find the nearest or cheapest available EV charging station.

5.4.1 The system shall retrieve EV charging stations based on the **User's** location or selected location.

5.4.2 The system shall recommend EV charging stations using the same preference-based recommendation logic (3. Preference-Based Recommendation).

5.4.3 The system shall display search results as a sorted list.

5.4.3.1 The system shall display details of the EV charging station as in 5.3.1

5.4.4 The system shall display a message when no EV charging stations are found.

5.5 The system shall allow **Users** to confirm and book a charging station.

5.5.1 The system shall send a booking confirmation upon booking of charger.

5.5.2 The system shall deduct $5 as booking fee from the user's wallet upon successful booking of a charging station.

5.5.2 The system shall reserve the booking for 30-minutes in which the **Users** must reach the charger and start charging (check-in/plug-in).

5.5.3 The system shall automatically mark the booking as no-show if the **Users** has not started the charging session by the end of the 30-minutes grace period.

5.5.4 The system shall automatically release the EV charger back to the pool of available chargers once a booking is marked as no-show after the grace period.

5.5.5 The system shall update the charger status in real time so that newly released chargers are immediately available for other **Users** to book.

5.5.6 The system shall notify the **Users** (e.g., in-app notification, push notification) when the grace period is about to end and when their booking is cancelled due to no-show.

5.6 The system shall block EV charger booking attempts if wallet balance is insufficient.

5.6.1 The system shall prompt **Users** to top up their wallet if there is insufficient funds in their wallet.

5.7 The system shall allow **Users** to cancel a confirmed booking 30 minutes before the reserved time

6. Routing and Navigation

Description:

The system shall generate navigation routes and provide real-time guidance.

Functional Requirements:

6.1 The system shall generate a route from the current location to a selected carpark or EV charger.

6.2 The system shall allow **Users** to start navigation.

6.2.1 The system shall update navigation information based on the **User's** movement.

6.2.2 The system shall allow the **Users** to stop navigation at any time.

7. Wallet & Payment (EV Booking)

Description:

The system shall provide a digital wallet for managing payments and booking fees for EV charging reservations.

Functional Requirements:

7.1 The system shall provide a "Profile" button on the navigation header that redirects **Users** to Profile page which displays the User's current wallet balance in local currency format.

7.2 The system shall provide a "Top Up" button in the base map that redirects the Users to the Top Up page.

7.2.1 The system shall display exactly four predefined top-up amounts on the Top Up page.

7.2.2 The system shall allow the user to select one top-up amount at a time.

7.2.3 The system shall provide another "Top Up" Button.

7.2.4 Upon clicking "Top Up", the system shall display exactly two payment methods: Card and PayNow.

7.2.5 If the user selects PayNow, the system shall generate a unique payment QR code for the current top-up transaction.

7.2.6 The generated PayNow QR code shall correspond to the user's selected top-up amount.

7.2.7 If the user selects Card, the system shall display the following mandatory fields: Card Number, CVV, and Expiry Date.

7.2.8 The system shall provide a "Pay" button to validate the payment through card or the to validate payment through "pay now" is done.

7.3 Upon receiving confirmation of a successful transaction from the payment gateway.

7.3.1 Display a "Top Up Successful" message.

7.3.2 Update the user's wallet balance in the database.

7.4 The system shall display an error message if the user violates the requirement in 7.3.

7.5 The system shall record booking and top up related transactions in a transaction history.

7.5.1 The system shall provide a "Transaction History" button that redirects the user to the Transaction History page.

7.5.1 The system shall display a chronological list of the 20 most recent Booking or Top Up transactions.

7.5.2 Each transaction shall display a mandatory field for the Top Up Amount, Date & Time, Invoice number and Transaction Type.

7.5.3 The system shall display a "No Transaction found" message if the user has no history.

2. Nonfunctional Requirements

2.1 Performance Requirements

2.1.1 General Performance

- The system shall provide responsive interactions for all core user actions which includes: map display, searching, filtering, navigation, booking and payment.
- The system shall ensure that user requests are processed within an acceptable delay under normal operating conditions.
- Real-time availability updates for carparks and EV chargers shall refresh every 60 seconds.
- The system shall update the route within 10 seconds of detecting traffic delays >5 minutes, road closures, or **Users** missing a stop (>100m past without turn).

2.1.2 Map and Location Services

- The system shall be able to load the interactive map within an acceptable delay after the application is launched.
- The system shall be able to display the nearby facilities: carparks, EV charging stations and petrol stations after an acceptable delay when a search or filter is applied.
- The system shall be able to calculate and display the navigation routes with a response time that supports a smooth and uninterrupted experience for the user.

2.1.3 Data Retrieval and Updates

- Facility information such as availability and pricing needs to be refreshed periodically to reflect current facility conditions.
- The system shall be able to support up to [10,000] concurrent users while maintaining acceptable performance levels.

2.2 Safety Requirements

- The system shall be able to display notifications or warnings when location, pricing or other variable data is unavailable or outdated.
- The system shall be able to allow the users to stop or exit navigation at any time.
- The system shall be able to maintain booking consistency under concurrent user access by rejecting conflicting booking requests and notifying the affected users.
- The system shall be able to recover from unexpected errors without crashing or losing user data.

2.3 Security Requirements

- The system shall require user authentication to access personal features such as profiles, wallet, bookings and transaction history.
- The system shall be able to protect user credentials with appropriate security measures.
- The system shall securely store user credentials using encryption.
- The system shall not be able to access or use the user’s location data without the user's consent.
- All API communications shall use HTTPS/TLS 1.3 with Perfect Forward Secrecy.

2.4 Software Quality Attributes

2.4.1 Usability

- The system shall provide a user interface that is easy to understand and use on mobile devices.
- The system shall provide clear guidance within the mobile application to support users with the following functions: searching, filtering, navigation and payment.

2.4.2 Reliability

- The system shall be able to remain operational during normal usage conditions.
- The system shall be able to troubleshoot and recover from failures regarding network connectivity or external factors.

2.4.3 Maintainability

- The system shall be designed to support future enhancements without affecting existing functionality.

- The system shall allow updates and fixes without major disruption to users.

2.5 Business Rules

- Only registered users shall be allowed to access personalized features such as wallet, bookings and transaction history.
- Users shall only be allowed to book EV charging slots that are available at the selected timeslot.
- Payments will only be processed when there is sufficient wallet balance.
- Users shall be allowed to cancel bookings subjected to cancellation terms and conditions.
- The system shall record completed transactions and make them available for users to review.

3. Data Dictionary

Term	Definition
User	An individual who interacts with the system as a registered user or yet to be registered user.
Registered Users	An individual with an account, whose information and records are in the system and has access to all the user features.
User Account	A registered profile in the system that stores a user's email address, password and profile information and enables access to features including EV charging slot booking, payment & transaction history.
Profile	A record that stores information about the user's name & handphone number.
Email Verification Link	A unique, time-limited URL sent to the user's email address provided during registration, which enables them to activate their account and become a registered user upon clicking.
Reset Link	A unique, time-limited URL sent to the user's registered email address that enables them to change their password to a new one.

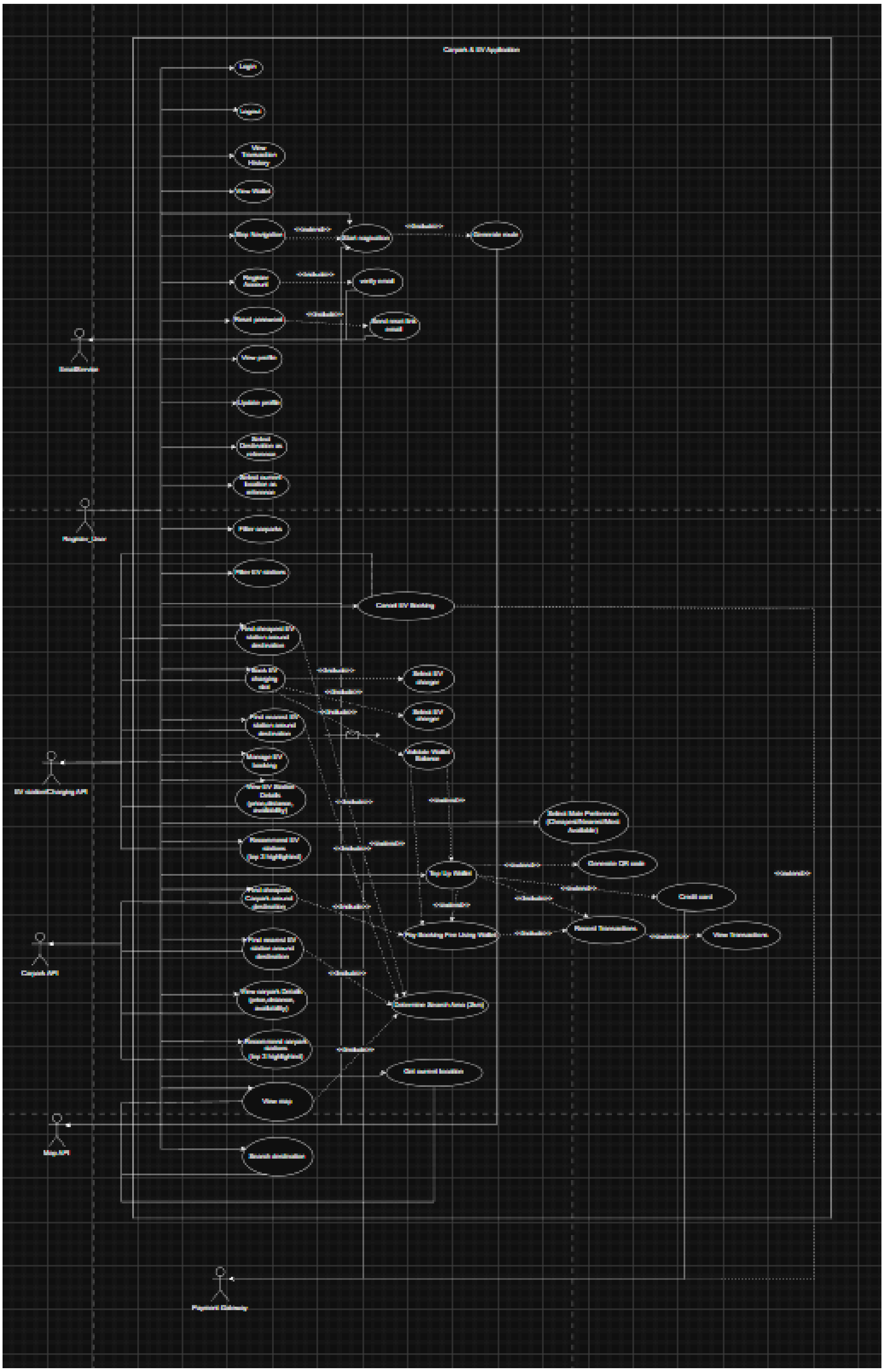
Current Location	An accurate reflection of the user's current real time geographic position determined by their device's geolocation services.
Destination	A specific location or address that is selected by the user on the map.
Search Radius	A displacement (in kilometers) from either the user's current location or selected destination within which the system searches for and displays carparks & EV charging stations. The default value is 2km.
Carpark	A venue that provides parking spaces for vehicles of various types. Carparks may or may not include EV charging stations.
Carpark Vacancy	A color-coded UI element that indicates real-time carpark availability. Green represents many slots available, Yellow represents limited slots, and Red represents a full/near full carpark.
EV	An abbreviated form for Electric Vehicle.
EV Charging Station	A designated area within or near a carpark that contains one or more charging points where electric vehicles can be charged.
EV Charging Point	An individual charging unit within an EV charging station that can charge one EV at a time. It accepts bookings from the users.
Adapter Type	A connector that is provided by an EV charging point which determines the compatibility between the charger and different EV models along with the charging speed.
EV Charging Provider	A company or organisation that owns and operates an EV charging station, managing its maintenance, pricing, and availability.
Booking	A reservation made by a registered user to secure an EV charging point from a specific time onwards, the charger is held exclusively for that user for 30 minutes before it is

	released back into the pool if user did not use the charging spot.
Booking Confirmation	A notification sent to the user after successfully reserving an EV charging point, containing the details of the reservation.
EV Availability Status	The real-time operational state of an EV charging point, indicating whether it is available for booking or occupied.
Booking Status	The current state of a user's EV booking, which can be Confirmed (waiting for check in), Checked-In(user has checked in), Cancelled(user cancelled during the grace period) or No Show (user did not show up during grace period).
Check-In	The action of a user physically arriving at their booking venue and successfully began charging their car within grace period.
Grace Period	A 30-minute window after the booking start time during which the user must arrive at the reserved EV charging point and begin charging (check-in). If the user fails to check in within this period, the booking is automatically marked as a no-show, the deposit is forfeited, and the charging point is released back to available inventory.
No-Show	A status applied to a booking when the user does not arrive and starts charging within the grace period, resulting in automatic cancellation, loss of deposit, and release of the reserved charging point.
Route	A calculated navigation path from the user's current location to a selected carpark or EV charging station
Wallet	A secure electronic account within the system where registered users can store funds for booking. Users can top up their money and view their transaction history.
Wallet Balance	An amount that reflects the sum of the user's funds after topping up, refunds, booking costs.

Booking Fee	A charge deducted from the user's wallet balance when an EV charging slot reservation is confirmed.
Supported Payment Methods	The accepted payment options that the software supports which users can use to deposit funds into their wallet.

4. Initial Use Case Model

4.1 Use Case Diagram



Use case diagram link:
https://app.diagrams.net/#G17RwjNEy5TaJhDc32ExDHNGTzh5xBfcPz#%7B%22pageId%22%3A%223g6re_2a6CxrNLDaZCts%22%7D

4.2 Use Case Description

Use Case ID:	1		
Use Case Name:	Login		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows a registered user to log in to the system by providing valid login details in order to access the system features.
Preconditions:	1. The user has a registered account. 2. The user is not currently logged in.
Postconditions:	1. The user is successfully authenticated and logged into the system 2. The system displays the home screen with the main map.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user launches the application. 2. The system displays the login screen. 3. The user enters their email/username and password. 4. The user submits the login request. 5. The system validates the entered details. 6. The system authenticates the user successfully. 7. The system shows home page with main map
Alternative Flows:	AF-1 1. At step 5, if a user enters wrong credentials, the system displays an error message indicating incorrect email/username or password. 2. The system prompts the user to re-enter their credentials. 3. The use case resumes at step 3. AF-2 1. At step 2, the user selects the “Forgot Password” option. 2. The system initiates the reset password use case.
Exceptions:	Network Error

	1. If network error happens during authentication, the system displays an error message saying that login cannot be completed. 2. Use case terminates.
Includes:	None
Special Requirements:	1. Password input must be masked
Assumptions:	1. User has stable internet connection
Notes and Issues:	None

Use Case ID:	2		
Use Case Name:	RegisterAccount		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	A new user registers for an account by providing the required registration details. The system validates the information and sends a verification email to the provided email before creating a new user account upon successful validation.
Preconditions:	1. The user is not logged in. 2. The registration screen is accessible.
Postconditions:	1. A new user is successfully created. 2. The user is able to log in using the registered credentials.
Priority:	High
Frequency of Use:	One-off
Flow of Events:	1. The guest selects the Register Account option. 2. The system displays the registration form. 3. The guest enters the required registration details. 4. The systems sends validation email to provided email 5. The system confirms the email of the new account. 6. The system creates a new user account. 7. A confirmation message is displayed to the guest.
Alternative Flows:	If the user enters invalid or incomplete information, the system shall display an error message and prompt the user to re-enter the information for that particular column.
Exceptions:	1. If registration service is unavailable, an error message would be displayed to inform the guest that registration cannot be completed at this time.
Includes:	None

Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	3		
Use Case Name:	Reset Password		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows users who do not remember their password to reset it by requesting a password reset and they will be able to set a new password of their choice.
Preconditions:	1. The User has a registered account 2. The user is not logged in
Postconditions:	1. The user’s password is successfully updated 2. The user is able to log in using the new password
Priority:	High
Frequency of Use:	Medium
Flow of Events:	1. The user selects the “Forgot Password” option on the login screen. 2. The system prompts the user to enter their registered email address. 3. The user enters their email address and submits the request. 4. The system verifies that the email address exists in the system. 5. The system sends a password reset link or verification code to the user’s email address. 6. The user accesses the reset link or enters the verification code. 7. The system prompts the user to enter a new password. 8. The user enters and confirms the new password. 9. The system updates the user’s password successfully. 10. The system displays a confirmation message indicating that the password has been reset.
Alternative Flows:	AF-1: Email address not registered 1. At step 4, if the entered email address is not registered then the system displays an error message.

	2. The system prompts the user to re-enter a valid email address. 3. The use case resumes at Step 2. AF-2: Invalid/ Expired Reset Link 1. At step 6, if the reset link/verification code is invalid or expired then the system will display an error message. 2. The system prompts the user to request a new password reset. 3. The use case resumes at Step 1.
Exceptions:	Network or Email service failure 1. If the system fails to send a resend link or verification code, the system displays an error message. 2. Use case terminates.
Includes:	None
Special Requirements:	1. The new password must comply with password strength requirements. 2. Password input must be masked.
Assumptions:	1. The user has access to the registered email account.
Notes and Issues:	None

Use Case ID:	4		
Use Case Name:	Send_Reset_Link_Email		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	Guest
Description:	The system sends a password reset link to the registered email address provided by the user after a password reset request is initiated.
Preconditions:	1. The user has selected the Reset Password option. 2. The user has provided a registered email address.
Postconditions:	1. A password reset link is sent to the user’s email address. 2. The user can use the link to reset their password.

Priority:	High
Frequency of Use:	High
Flow of Events:	1. The guest submits a registered email address for password reset. 2. The system generates a unique password reset link. 3. The system sends the reset link to the user's email address. 4. The system sends the reset link to the user’s email address. 5. A confirmation message is displayed to the guest.
Alternative Flows:	If the email address is not registered, the system would display an error message and prompt the guest to re-enter a valid registered email address.
Exceptions:	1. If Email Service is unavailable, an error message would be displayed to inform the guest that the reset email is unable to be sent at this time.
Includes:	Extends: Reset Password
Special Requirements:	The reset link must expire after a fixed time period for security purposes.
Assumptions:	1. Network connection is available. 2. The provided email address is accessible by the user.
Notes and Issues:	None

Use Case ID:	5		
Use Case Name:	Manage_EV_Booking		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user views and manages their existing EV charging bookings. The user can review booking details and have the option to cancel the EV charging booking.
Preconditions:	1. The user must be logged in. 2. The user must have at least one existing EV charging booking.
Postconditions:	1. The user’s EV charging bookings are displayed. 2. Any selected booking is updated or cancelled accordingly.
Priority:	High
Frequency of Use:	Use-based
Flow of Events:	1. The user selected the **Manage EV Booking** option.

	2. The system retrieves the user’s existing EV charging bookings. 3. The user chooses the EV charging booking to cancel. 4. The system updates the booking status and displays a confirmation message.
Alternative Flows:	If the user chooses not to cancel the booking, the system shall return to the home screen with main map.
Exceptions:	1. If booking data cannot be retrieved, an error message is displayed to inform the user that the EV booking services are currently unavailable.
Includes:	Cancel EV Charging Booking
Special Requirements:	Booking updates must be reelected in real time.
Assumptions:	1. Network connection is available. 2. EV booking data is stored and accessible.
Notes and Issues:	None

Use Case ID:	6		
Use Case Name:	Stop Navigation		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows the user to stop an currently ongoing navigation and return to the main map screen.
Preconditions:	1. The user has started navigation to a selected carpark of EV charger 2. Currently, the user is running a navigation session.
Postconditions:	1. Navigation Stopped 2. Systems returns the user to the home screen with the main map.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user is viewing the active navigation screen. 2. The user taps the “Stop Navigation” button. 3. The system prompts the user to confirm stopping navigation. 4. The user confirms the action. 5. The system stops navigation and ends route updates. 6. The system returns the user to the main map screen at the current location.

Alternative Flows:	AF-1: User cancels stop navigation 1. At Step 4, if the user selects “Cancel”, the system keeps navigation running. 2. The use case terminates.
Exceptions:	None
Includes:	None
Special Requirements:	None
Assumptions:	None
Notes and Issues:	None

Use Case ID:	7		
Use Case Name:	Recommended_Nearest_Carpark		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user finds the nearest available carpark based on a selected reference location. The system evaluates nearby carparks using various factors and identifies the best shortest distance from the reference point.
Preconditions:	The user is on the map interface.
Postconditions:	A reference location (current location or selected destination) is available.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selected the **Find Nearest Carpark** option. 2. The system retrieves the reference location. 3. The system retrieves the nearby carpark data from the carpark API. 4. The system will use our recommendation logic to recommend the nearest carpark 5. The nearest car park information is displayed to the user.
Alternative Flows:	If no reference location is available, the system prompts the user to select a reference location.
Exceptions:	1. If carpark data cannot be retrieved, an error message will be displayed to inform the user that the nearest carpark services are currently unavailable.
Includes:	Determine Search Area Evaluate Options

Special Requirements:	Distance calculations must be accurate within an acceptable tolerance.
Assumptions:	1. Network connection is available. 2. Location data is accurate and accessible.
Notes and Issues:	None

Use Case ID:	8		
Use Case Name:	Recommended Cheapest Carpark		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows user to find the best cheapest carpark with system's algorithm near their selected starting point
Preconditions:	1. User provides either current location or selected destination 2. System retrieves a list of nearby carparks within the search area 3. Capark pricing data is available.
Postconditions:	1. The system displays carparks ranked by lowest prices (cheapest first) 2. The cheapest carpark(s) is highlighted on the list for the user.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user is on the map screen with nearby carparks displayed. 2. The user selects the "Cheapest" option 3. The system uses our recommendation logic to recommend the cheapest carpark 4. The system highlights the cheapest carpark and displays the ranked results 5. The user selects a carpark to view details (pricing, availability) or to route to it.
Alternative Flows:	AF-1: Some Carparks Have Missing Pricing 1. At step 3, if some carparks do not have pricing data, the system labels those carparks as "Price unavailable". 2. The system ranks only the carparks with publicly available known pricing data.

	3. The use case continues at Step 5.
Exceptions:	EX-1: No Pricing Data Available 1. If the system cannot retrieve pricing data for all nearby carparks, the system displays an error message indicating pricing is unavailable. 2. The system displays nearby carparks without ranking by price. 3. The use case terminates.
Includes:	None
Special Requirements:	None
Assumptions:	1. Carpark pricing data received by the system is accurate and up to date
Notes and Issues:	None

Use Case ID:	9		
Use Case Name:	LogOut		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user logs out of the system to terminate the current authenticated session and return to a non-authenticated state.
Preconditions:	1. The user is logged in.
Postconditions:	1. The user session is terminated. 2. The user is redirected to the login or home screen.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the Logout option. 2. The system terminates the user’s active session. 3. The system clears any session-related data. 4. The system redirects the user to the login or home screen.
Alternative Flows:	None
Exceptions:	1. If session termination fails, an error message is displayed to inform the user that the logout was unsuccessful.
Includes:	None
Special Requirements:	Session data must be securely cleared upon logout.

Assumptions:	1. Network connection is available.
Notes and Issues:	None

Use Case ID:	10		
Use Case Name:	View Profile		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows a logged in user to view profile information and account details
Preconditions:	The user is logged in
Postconditions:	The user's profile information is displayed on the profile screen
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user is on the home screen with the main map 2. The user selects the "Profile" option (e.g., from the menu or profile icon). 3. The system retrieves the user's profile information. 4. The system displays the profile screen with the user's details (e.g., name, email/username, car type and model, email address)
Alternative Flows:	None
Exceptions:	EX-1: Unable to Load Profile 1. If the system fails to retrieve profile information due to a system or network error, the system displays an error message and prompts the user to retry. 2. The use case terminates.
Includes:	None
Special Requirements:	1. The profile screen must not display sensitive information (e.g. full password)
Assumptions:	None
Notes and Issues:	None

Use Case ID:	11

Use Case Name:	Update_Profile		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user updates their personal profile information. The system would validate and save the updated profile details.
Preconditions:	1. The user is logged in. 2. The user is on the profile management screen.
Postconditions:	1. The user's profile information updated. 2. The updated profile details are saved in the system.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user selects the Update Profile. 2. The system displays the user's current profile information. 3. The user modifies one or more profile fields. 4. The user submits the updated profile information. 5. The system validates the updated information. 6. The system saves the updated profile details. 7. A confirmation message is displayed to the user.
Alternative Flows:	If the updated information is invalid or incomplete, the system would display an error message and prompt the user to re-enter with correct details.
Exceptions:	1. If profile update service is unavailable, an error message would be displayed to inform the user that the profile update cannot be completed at this time.
Includes:	None
Special Requirements:	Updated profile data must be saved securely.
Assumptions:	1. Network connection is available.
Notes and Issues:	None

Use Case ID:	12		
Use Case Name:	View EV charger details		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User

Description:	This use case allows the user to view the real time EV charging stations details such as availability, price & distance. The user can click on a specific EV charging station to view its full details and availability.
Preconditions:	1. Either current location or selected destination is available 2. The system retrieves a list of nearby EV charging stations
Postconditions:	1. The details of EV chargers is displayed to the user.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user is on the map screen with nearby EV charging stations displayed. 2. The user selects an EV charging station from the map or list. 3. The system retrieves real-time availability, price and distance information for the selected EV charger. 4. The system displays the availability details (e.g., number of available chargers, occupied chargers, and status), price and distance from destination
Alternative Flows:	AF-1: Availability Information Partially Available 1. At step 3, if real-time information is partially available, the system displays the available data and labels missing information as “Unavailable”. 2. The use case continues at step 4.
Exceptions:	None
Includes:	None
Special Requirements:	Availability information should be refreshed periodically so that the latest EV charger information is displayed.
Assumptions:	EV charger availability data is accurate and up to date
Notes and Issues:	None

Use Case ID:	13		
Use Case Name:	View Carpark details		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User

Description:	The user views the availability of parking lots, pricing and distance from destination to selected carpark. The system displays real-time availability information to help the user make a parking decision.
Preconditions:	1. The user is on the map interface. 2. A carpark has been selected by the user.
Postconditions:	1. Th information of the selected carpark is displayed.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects a carpark on the map. 2. The system retrieves data from the carpark API for the selected carpark. 3. The system displays the number of available parking lots, pricing and distance to selected carpark
Alternative Flows:	If availability data cannot be retrieved, the system displays a message indicating that availability information is unavailable.
Exceptions:	1. If carpark information is unavailable, an error message will be displayed to inform the user that the carpark information cannot be retrieved at this time.
Includes:	None
Special Requirements:	Availability data should be updated regularly to reflect real-time conditions.
Assumptions:	1. Network connection is available 2. Carpark availability data is provided by the Carpark API and is accessible by the system.
Notes and Issues:	None

Use Case ID:	14		
Use Case Name:	View Carpark Pricing		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	This use case allows the user to view the pricing details of a selected carpark
Preconditions:	1. Either current location or selected destination is available 2. Nearby carparks are displayed first on the map 3. Pricing data is available for the displayed carparks.
Postconditions:	1. The selected carpark's pricing details are shown to the user
Priority:	High

Frequency of Use:	High
Flow of Events:	1. The user is on the map screen with the nearby carparks displayed 2. The user taps on a carpark pink or selects a carpark from the the list 3. The system displays the carpark details panel 4. The pricing is shown directly in the details panel
Alternative Flows:	AF-1: Pricing Data Partially Available 1. At Step 5, if only some pricing information is available, the system displays the available pricing details and labels missing fields as “Unavailable”. 2. The use case continues at Step 6.
Exceptions:	EX-1: Pricing Data Not Available 1. If the system cannot retrieve pricing data for the selected carpark, the system displays a message such as “Pricing information is unavailable”. 2. The system still displays other carpark details (e.g., location, availability if available). 3. The use case terminates.
Includes:	Record Transactions
Special Requirements:	1. Pricing must clearly state the pricing basis (per house/ per entry/ free)
Assumptions:	1. Carpark pricing data is accurate and up to date.
Notes and Issues:	None

Use Case ID:	15		
Use Case Name:	View_EV_Station_Locations		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user views the locations of EV charging stations on the map. The system displays available EV stations to help the user locate nearby charging facilities.
Preconditions:	1. The user is on the map interface.
Postconditions:	1. EV charging stations are displayed on the map.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user is on the home screen with the main map

	2. The user clicks on EV Charging option 3. The system retrieves the EV Charging location data within the search area 4. The system displays EV charging stations markers on the map 5. The user may tap a marker to view basic EV charging station details
Alternative Flows:	EX-1: No EV charging stations found 1. Systems displays a message indicating that no stations are available nearby 2. The use case terminates.
Exceptions:	1. If EV station service is unavailable, an error message will be displayed to inform the user that the EV station locations cannot be retrieved at this time.
Includes:	None
Special Requirements:	EV station locations should be displayed accurately on the map.
Assumptions:	1. Network connection is available. 2. EV station location data is provided by the EV Station API and is accessible by the system.
Notes and Issues:	None

Use Case ID:	16		
Use Case Name:	Record_Transactions		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user is able to view the history of their recorded EV charging bookings and wallet top ups.
Preconditions:	1. The user is logged in. 2. The user has at least one booking transaction or top up transaction
Postconditions:	1. The user's transaction history is displayed.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user selects the **View Booking Transactions** option. 2. The system retrieves the user's transaction records. 3. The system displays the transaction details to the user.

Alternative Flows:	1. If no transactions are found, the system would display a message to indicate that no booking transactions are available.
Exceptions:	1. If transaction data cannot be retrieved, an error message would be displayed to inform the s are currently unavailable. 2. The use case terminates.
Includes:	Extends: View Transactions
Special Requirements:	None
Assumptions:	1. Network connection is available. 2. Transaction data is stored and can be accessed by the system.
Notes and Issues:	None

Use Case ID:	17		
Use Case Name:	Start_Navigation		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user starts navigation to a selected destination. The system would provide route guidance and update the route information in real-time based on traffic and location changes.
Preconditions:	1. The user is on the map interface. 2. A destination(e.g. Carpark or EV charging station) has been selected. 3. Location services are enabled.
Postconditions:	1. Navigation to the selected destination is initiated. 2. Route information is continuously updated and displayed to the user.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the **Start Navigation** option. 2. The system retrieves the user's current location. 3. The system generates the optimal route to the selected destination. 4. The system displays navigation instructions to the user. 5. The system updates the route information in real-time as the user moves.
Alternative Flows:	1. If the location services are disabled, the system would prompt the user to enable location services.

Exceptions:	1. If the route data cannot be retrieved, an error message would be displayed to inform the user that navigation services are currently unavailable.
Includes:	Get Current Location Generate Route
Special Requirements:	Route updates must reflect real-time traffic conditions where available.
Assumptions:	1. Network connection is available. 2. Route and navigation data are provided by the Navigation API and can be accessed by the system.
Notes and Issues:	None

Use Case ID:	18		
Use Case Name:	Book_EV_ChargingSlot		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user books an EV charging slot by selecting an EV charging station and an available time slot. The system validates the booking details, processes the payment and confirms the booking upon successful payment.
Preconditions:	1. The user is logged in. 2. An EV charging station has been selected. 3. Available charging slots exist for the selected station. 4. The user has an active wallet account with sufficient balance.
Postconditions:	1. An EV charging slot has been successfully booked. 2. The booking details are stored in the system. 3. A booking confirmation is displayed to the user. 4. The payment transaction is recorded
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the **Book EV Charging Slot** option. 2. The system displays available EV charging stations. 3. The user selects an EV charging station 4. The system validates the selected booking details. 5. The system includes the Pay Booking Fee using Wallet use case 6. The system processes the booking payment.

	7. The system confirms the booking and displays the booking details to the user.
Alternative Flows:	1. If the user does not have sufficient wallet balance, the system would display an insufficient balance message and inform the user to top up their wallet. 2. The booking process is then terminated,
Exceptions:	1. If the booking service is unavailable, an error message would be displayed to inform the user that the EV charging booking cannot be completed at this time.
Includes:	Selects EV Charger and Timeslot Pay Booking Fee Using Wallet Validate Wallet Balance
Special Requirements:	1. Booking confirmation must be generated immediately after successful payment.
Assumptions:	1. Network connection is available. 2. EV charging availability data is provided by the EV Charging API and can be accessed by the system.
Notes and Issues:	None

Use Case ID:	19		
Use Case Name:	Search_Destination		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user searches for a destination by entering a location or place name or by selecting the current location as the destination. The system would retrieve matching locations and display the selected destination on the map.
Preconditions:	1. The user is on the map interface. 2. Network connection is available. 3. The user is either entering a destination name or using the current location as a reference point.
Postconditions:	1. The selected destination is displayed on the map. 2. The destination is set as the reference location for further actions.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the destination search bar. 2. The system displays a text input field 3. The user enters a destination or place name or presses the current location icon.

	4. The system retrieves matching destination results. 5. The user selects a destination from the search results. 6. The system displays the selected destination on the map.
Alternative Flows:	1. If no matching destinations are found, the system would display a message to indicate that no results are available.
Exceptions:	1. If the destination search service is unavailable, an error message would be displayed to inform the user that the destination search cannot be completed at this time.
Includes:	None
Special Requirements:	1. Search results should be displayed promptly to ensure usability.
Assumptions:	1. Network connection is available. 2. Destination search data is provided by the Location Search API and can be accessed by the system.
Notes and Issues:	None

Use Case ID:	20		
Use Case Name:	TopUp_Wallet		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user tops up their wallet by adding funds to increase the wallet balance. The system processes the top-up request and updates the wallet balance upon successful completion.
Preconditions:	1. The wallet balance is increased by the selected top-up amount. 2. The top-up transaction is recorded in the system.
Postconditions:	1. The wallet balance is increased by the selected top-up amount. 2. The top-up transaction is recorded in the system.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the **Top Up Wallet** option. 2. The system displays available top-up amounts or allows manual input. 3. The user selects or enters a top-up amount. 4. The user confirms the top-up request.

	5. The system processes the top-up transaction. 6. The system updates the wallet balance. 7. A top-up confirmation message is displayed to the user.
Alternative Flows:	1. If the user enters an invalid top-up amount, the system would display an error message to prompt the user to re-enter a valid amount.
Exceptions:	1. If the wallet top-up service is unavailable, an error message would be displayed to inform the user that wallet top-up cannot be completed at this time.
Includes:	Extends: Generate QR code Credit Card
Special Requirements:	1. Wallet balance updates must be reflected immediately after a successful top-up.
Assumptions:	1. Network connection is available. 2. Wallet transaction data is stored and accessible by the system
Notes and Issues:	None

Use Case ID:	21		
Use Case Name:	Select_Destination_as_Reference		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user selects a destination to be used as a reference location. The system sets the selected destination as the reference point for displaying nearby facilities and performing location-based searches.
Preconditions:	1. The user is on the map interface. 2. A destination has been searched or displayed on the map.
Postconditions:	1. The selected destination is set as the reference location. 2. Subsequent location-based features use the selected destination as reference.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects a destination from the map or search results. 2. The system confirms the destination selection. 3. The system sets the selected destination as the reference location.

	4. The system updates the map based on the selected reference location.
Alternative Flows:	1. If the user cancels the destination selection, the system will retain the previous reference location.
Exceptions:	1. If the destination reference cannot be set, an error message would be displayed to inform the user that the destination cannot be used as a reference.
Includes:	None
Special Requirements:	1. Reference location updates should occur immediately after selection.
Assumptions:	1. Network connection is available. 2. Destination data is provided by the Location Search API and is accessible by the system.
Notes and Issues:	None

Use Case ID:	22		
Use Case Name:	Use_Current_Location_As_Reference		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user uses their current location as the reference point. The system retrieves the user's current location and sets it as the reference location for location-based features.
Preconditions:	1. The user is on the map interface. 2. Location services are enabled on the user's device.
Postconditions:	1. The user's current location is set as the reference location. 2. Subsequent location-based features use the current location as reference.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user selects the **Use Current Location as Reference** option. 2. The system retrieves the user's current location. 3. The system sets the current location as the reference location. 4. The system updates the map based on the selected reference location.
Alternative Flows:	1. If the location services are disabled, the system would prompt the user to enable location services.

Exceptions:	1. If the Current location cannot be retrieved, an error message would be displayed to inform the user that the current location cannot be used as a reference.
Includes:	Get Current Location
Special Requirements:	1. Location retrieval should be accurate within acceptable tolerance.
Assumptions:	1. Network connection is available. 2. Current location data is provided by the device's location service and is accessible by the system.
Notes and Issues:	None

Use Case ID:	23		
Use Case Name:	View_Map		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User
Description:	The user views the interactive map interface. The system displays a map that allows the user to view locations, pan, zoom, and interact with map-based features.
Preconditions:	1. The application is launched. 2. Network connection is available.
Postconditions:	1. The interactive map is displayed to the user.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The user opens the application. 2. The system loads the map interface. 3. The system displays the interactive map to the user. 4. The user can pan and zoom the map.
Alternative Flows:	1. If the map cannot be loaded within an acceptable time, the system would display a loading or retry message.
Exceptions:	1. If the Map service is unavailable, an error message would be displayed to inform the user that the map cannot be displayed at this time.
Includes:	None
Special Requirements:	1. The map should support standard gestures such as pan and zoom.
Assumptions:	1. Network connection is available. 2. Map data is provided by the Map API and is accessible by the system.
Notes and Issues:	None

Use Case ID:	24		
Use Case Name:	Determine Search Area		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	System
Description:	This use case allows the system to determine the search area (eg within a 2km radius) around a reference location to search for nearby carparks or EV charging stations
Preconditions:	A reference location is available
Postconditions:	1. The search area is determined around the reference location(eg 2km radius) 2. The system is ready to retrieve nearby carparks or EV stations within the search area.
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user selects or sets a reference location 2. The system calculates the search area based on a predefined radius (eg 2km) around the reference location 3. The system stores the search area and prepares to fetch nearby facilities within the area
Alternative Flows:	None
Exceptions:	EX-1: Invalid Location Data 1. If the system cannot retrieve valid coordinates for the reference location, the system displays an error message. 2. The system prompts the user to select a valid reference location. 3. The use case terminates
Includes:	None
Special Requirements:	The search area must be calculated accurately based on the reference location
Assumptions:	The system has a predefined search radius (eg 2km)
Notes and Issues:	None

Use Case ID:	25
Use Case Name:	Get_Current_Location

Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	Map API
Description:	Retrieves the user;s current geographical location from the device’s location services from the Map API. The retrieved location is used internally by the system as a reference point for determining search areas and generating navigation routes.
Preconditions:	1. The application is running. 2. The user has granted location permission. 3. The device supports location services.
Postconditions:	1. The user’s current location coordinates are retrieved successfully. 2. The current location is stored temporarily as the system reference location.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system requests the current location from the Map API. 2. The Map API retrieves the device’s current location coordinates. 3. The system receives and stores the location data to process the data for uses such as route navigation.
Alternative Flows:	1. If the location cannot be retrieved, the system retries the request or returns a failure status to the calling use case.
Exceptions:	1. If the Map API fails to retrieve the current location, the system returns an error and location-dependent features cannot proceed.
Includes:	None
Special Requirements:	1. Location accuracy must be sufficient for map display and distance calculation. 2. Response time must be within acceptable limits to avoid blocking dependent cases.
Assumptions:	1. The Map API is available. 2. The device’s GPS functionality is operational.
Notes and Issues:	None

Use Case ID:	26
Use Case Name:	Generate_QRcode

Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

Actor:	User & System
Description:	This use case allows the system to generate a QR code for the user for transaction purposes.
Preconditions:	1. The user has completed a relevant booking or relevant transaction
Postconditions:	1. A QR is successfully generated for the user 2. The QR code is displayed to the user for further action
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user initiates a booking or top up wallet that requires a QR code 2. The system creates the QR code image 3. The system displays the QR code to the user, allowing them to scan, save, or print it
Alternative Flows:	QR Code Generation Failure 1. If the QR code generation fails, the system displays an error message indicating that the QR code could not be created
Exceptions:	None
Includes:	None
Special Requirements:	QR code must be scannable
Assumptions:	None
Notes and Issues:	None

Use Case ID:	27		
Use Case Name:	Pay using Credit Card		
Created By:		Last Updated By:	
Date Created:		Date Last Updated:	

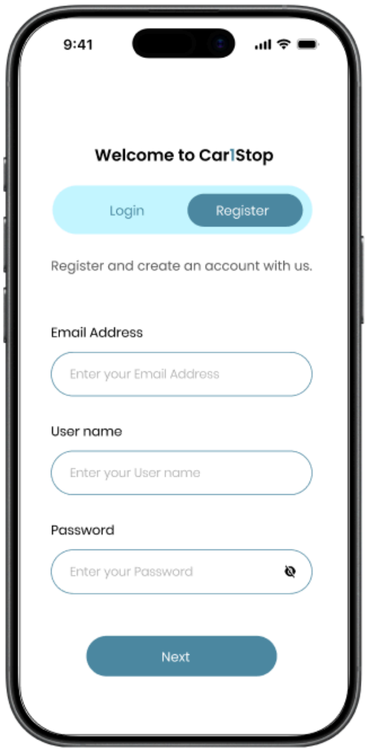
Actor:	User
Description:	This use case allows the user to make a payment for EV booking services via crediting after selecting a charger and confirming the details.
Preconditions:	1. User has selected an EV charger 2. User has a valid credit card linked to their account 3. User has internet connection

Postconditions:	1. Payment is successfully processed and the user is notified of the successful payment 2. User receives a confirmation of the payment transaction
Priority:	Medium
Frequency of Use:	Medium
Flow of Events:	1. The user selected EV charging station 2. User chooses credit card as the payment option 3. The system displays the payment screen with credit card details 4. The user enters their credit card information 5. The system securely processes the payment request via a third party payment 6. User receives a receipt and is notified that their payment was successful
Alternative Flows:	AF-1 Payment Declined - If credit card payment is declined, the system displays a payment failure message and prompts them to try another payment method AF-2 Payment Timeout - If the payment process takes too long, the system displays a timeout error and suggests the user to try another payment method
Exceptions:	None
Includes:	None
Special Requirements:	Credit Card information should be securely handled
Assumptions:	None
Notes and Issues:	None

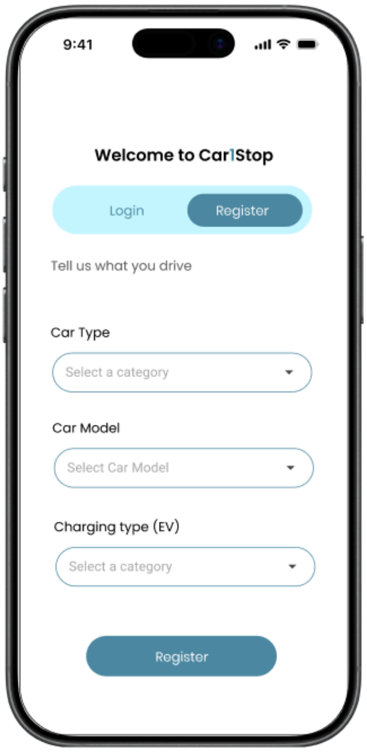
5. UI Mock Up

5.1 Registration, Validate and Login

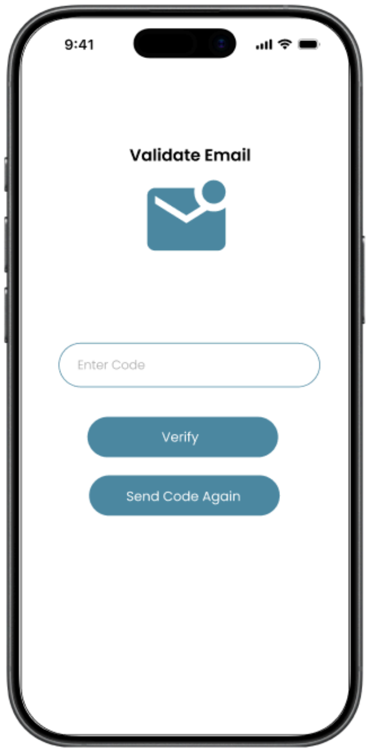
Register



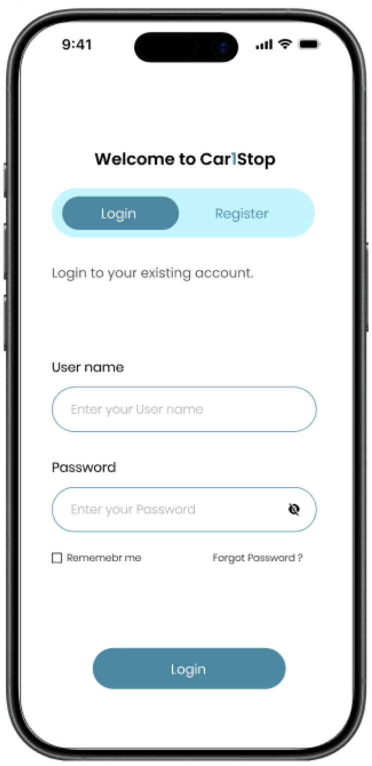
Register



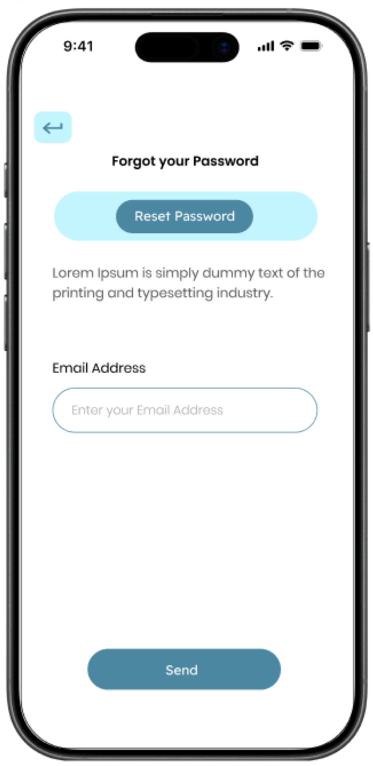
Register



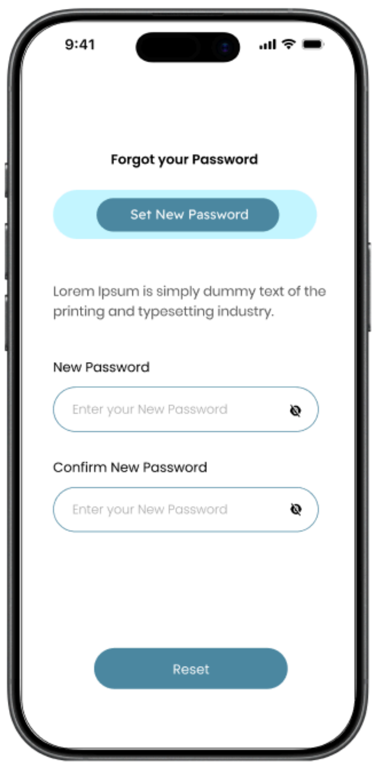
Login



Forget Password

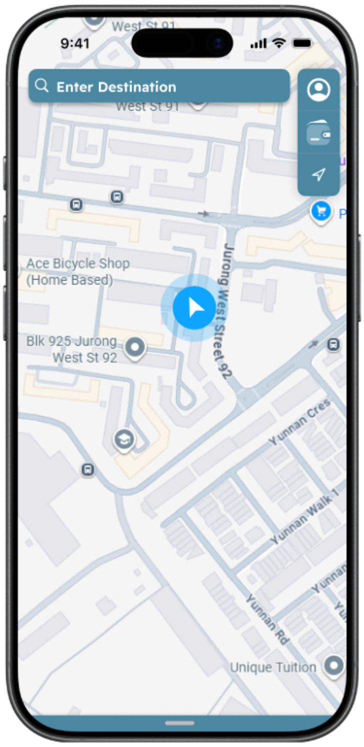


Reset Password

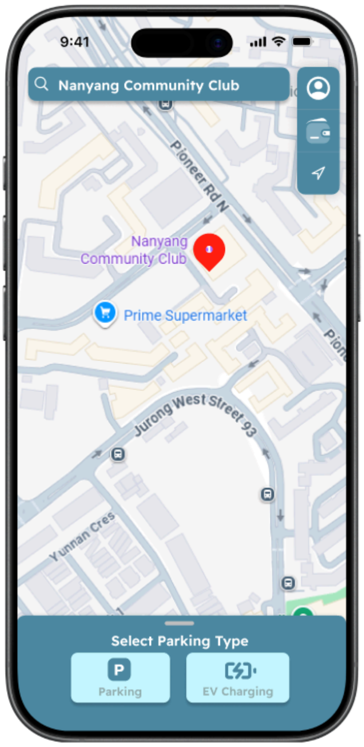


5.2 Base Map View

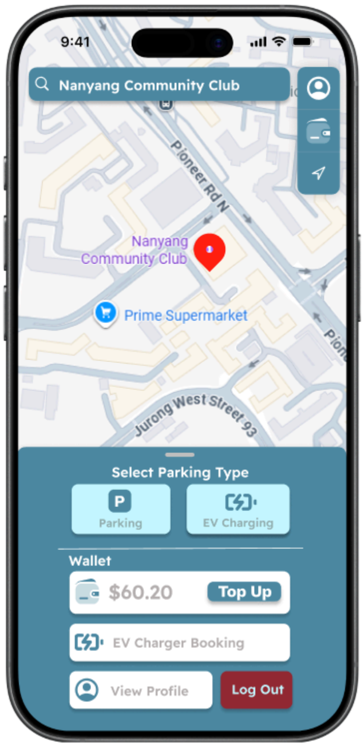
Base Map



Enter Destination

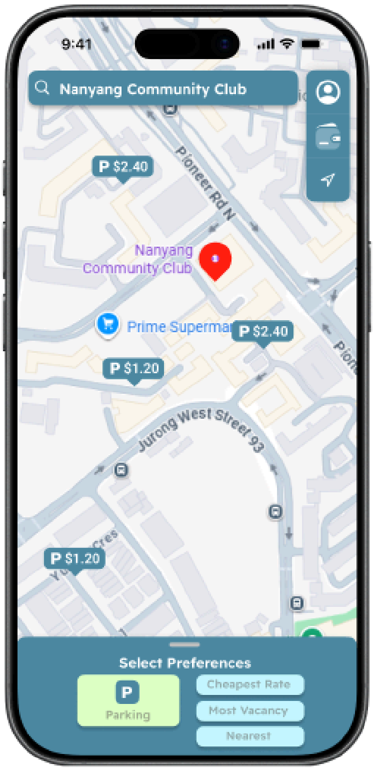


More Features

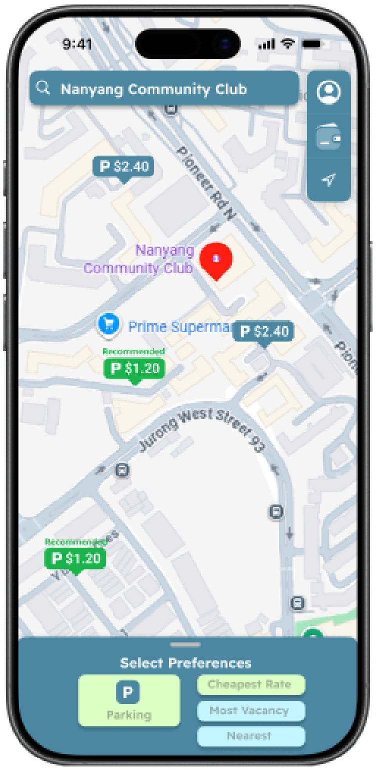


5.3 Carpark Finder, Preference Select and Recommendation

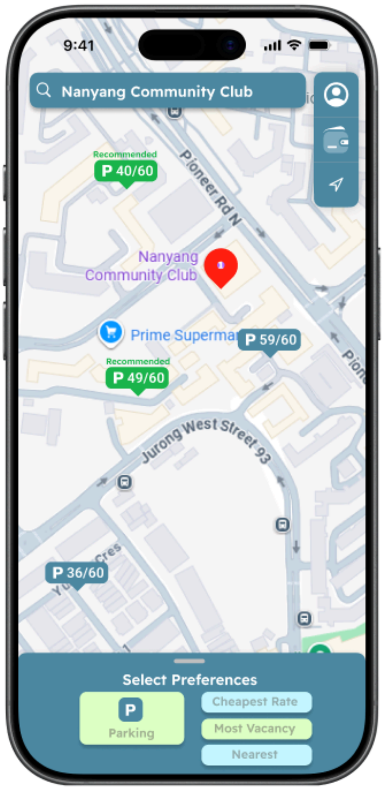
Select Parking Type



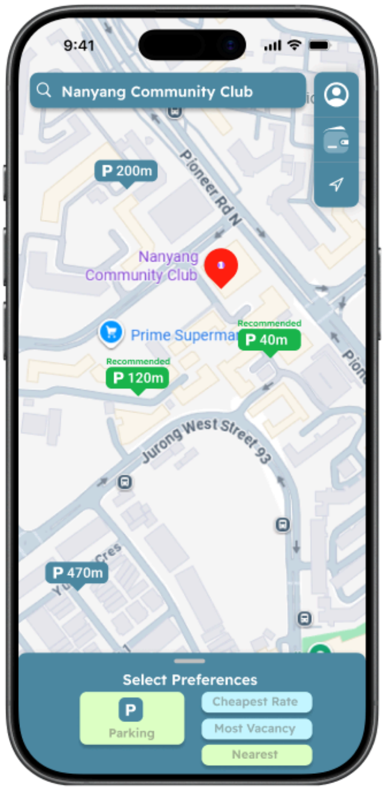
Select Cheapest Rate



Select Most Vacancy

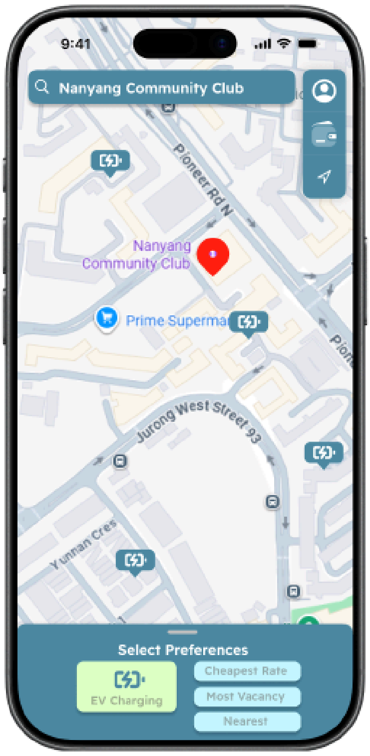


Select Nearest

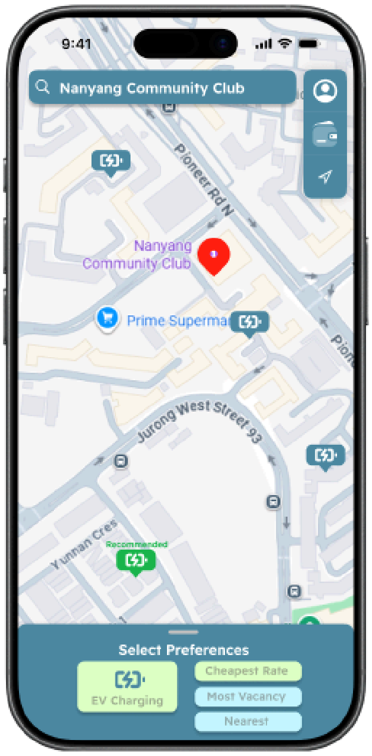


5.4 EV Charging Finder, Preference Select and Recommendation

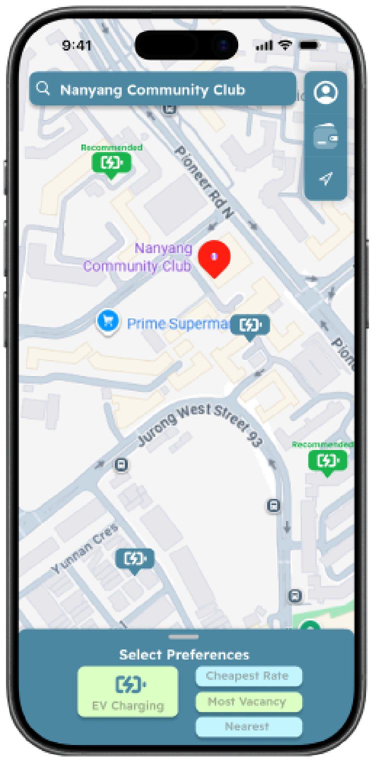
Select Parking Type



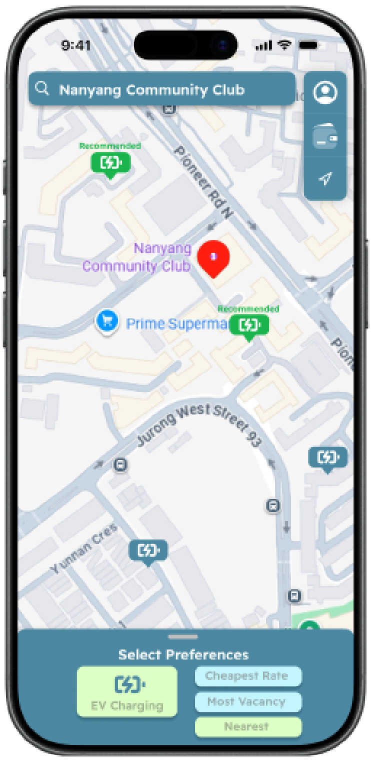
Select Cheapest Rate



Select Most Vacancy

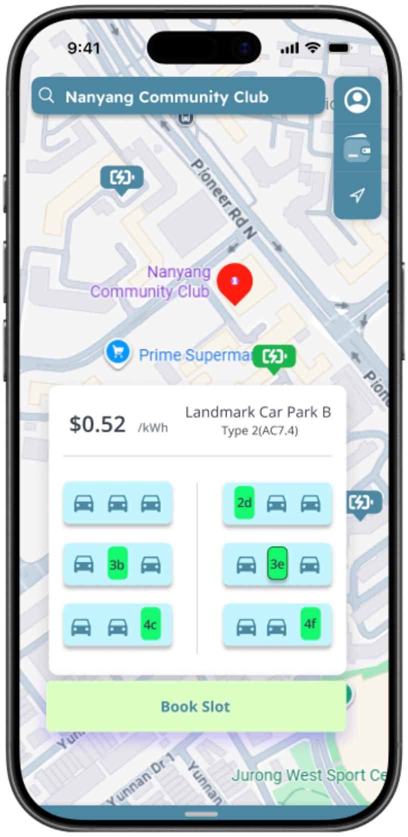


Select Nearest

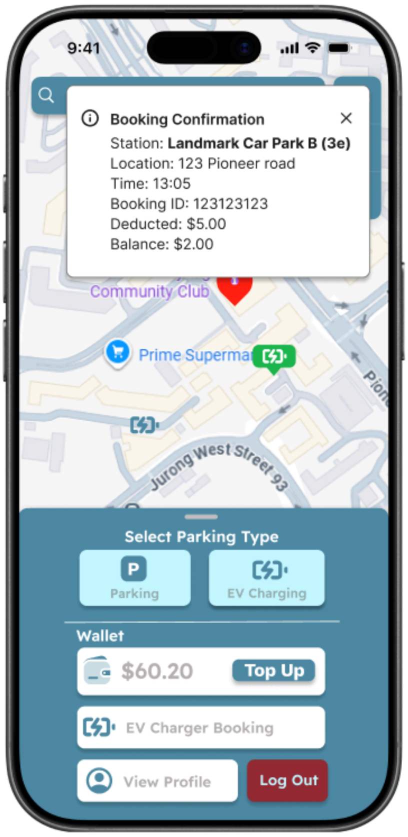


5.5 EV Booking, View Booking and Cancel Booking

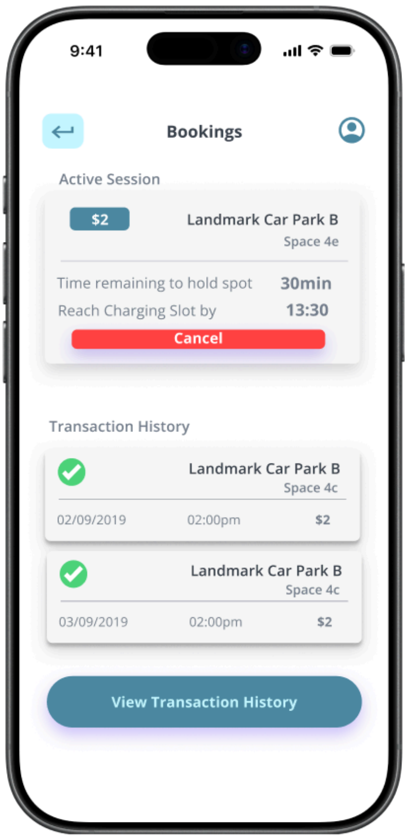
Select For Booking



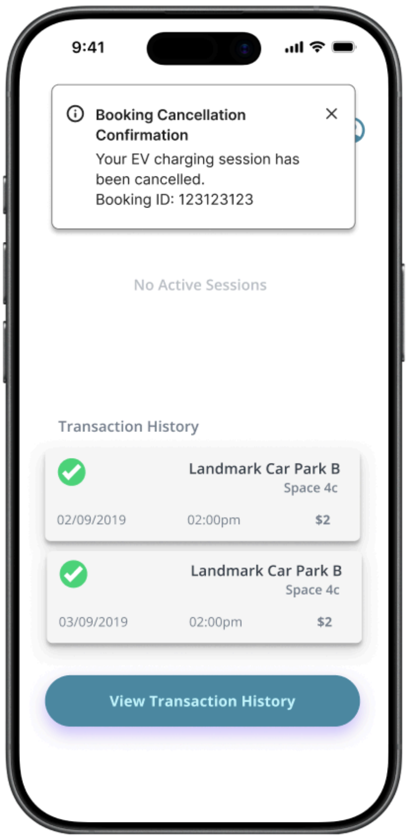
Booked EV Slot



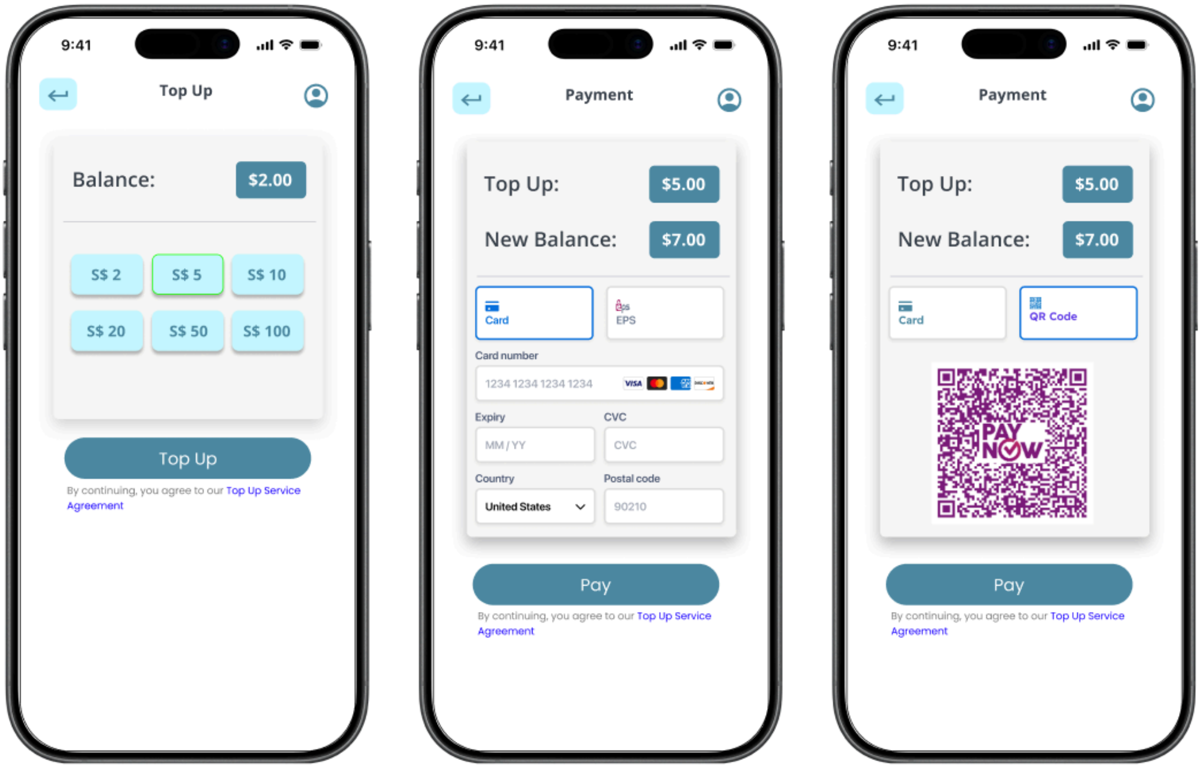
Ev Booking



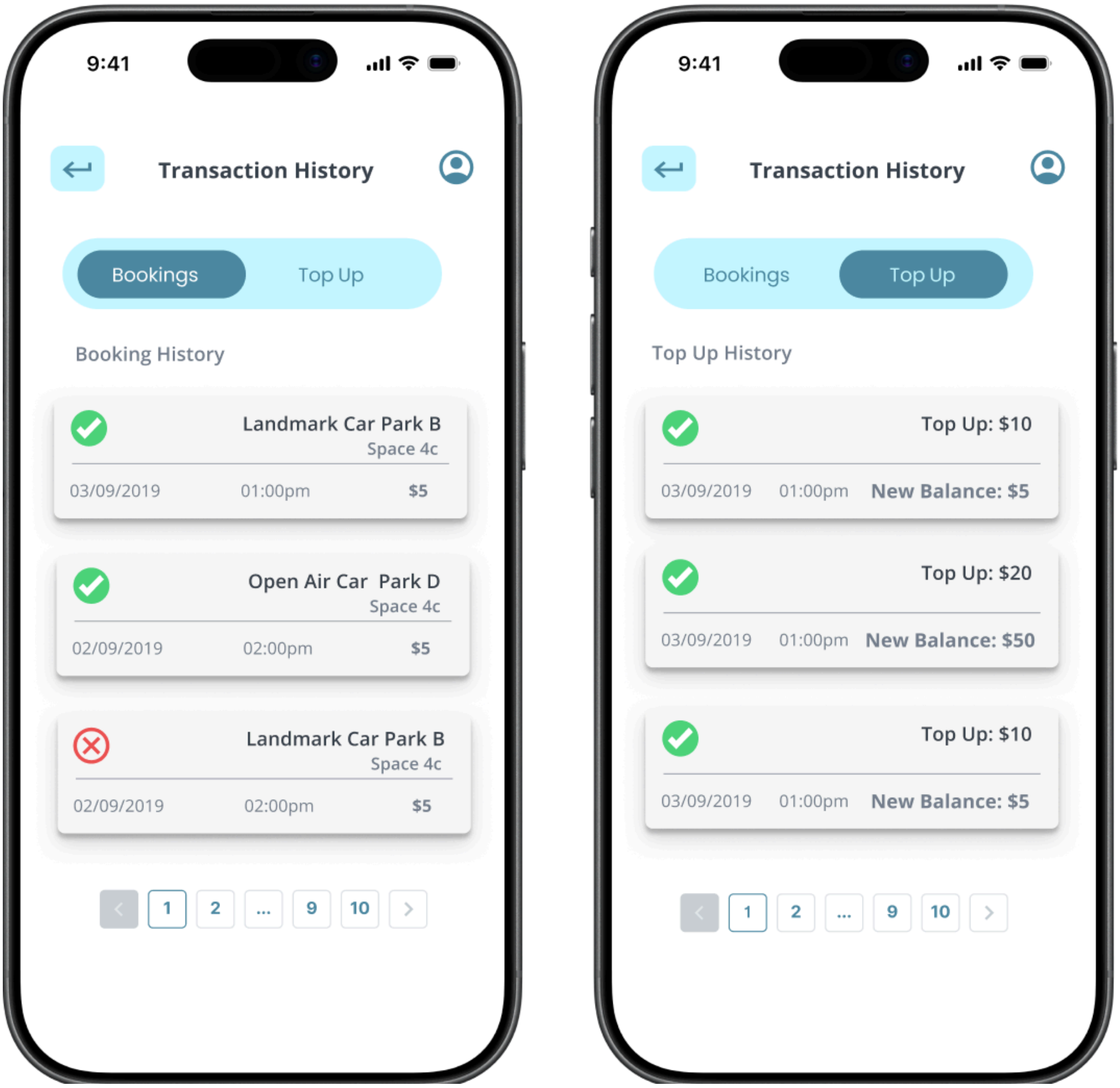
Cancelled Booking



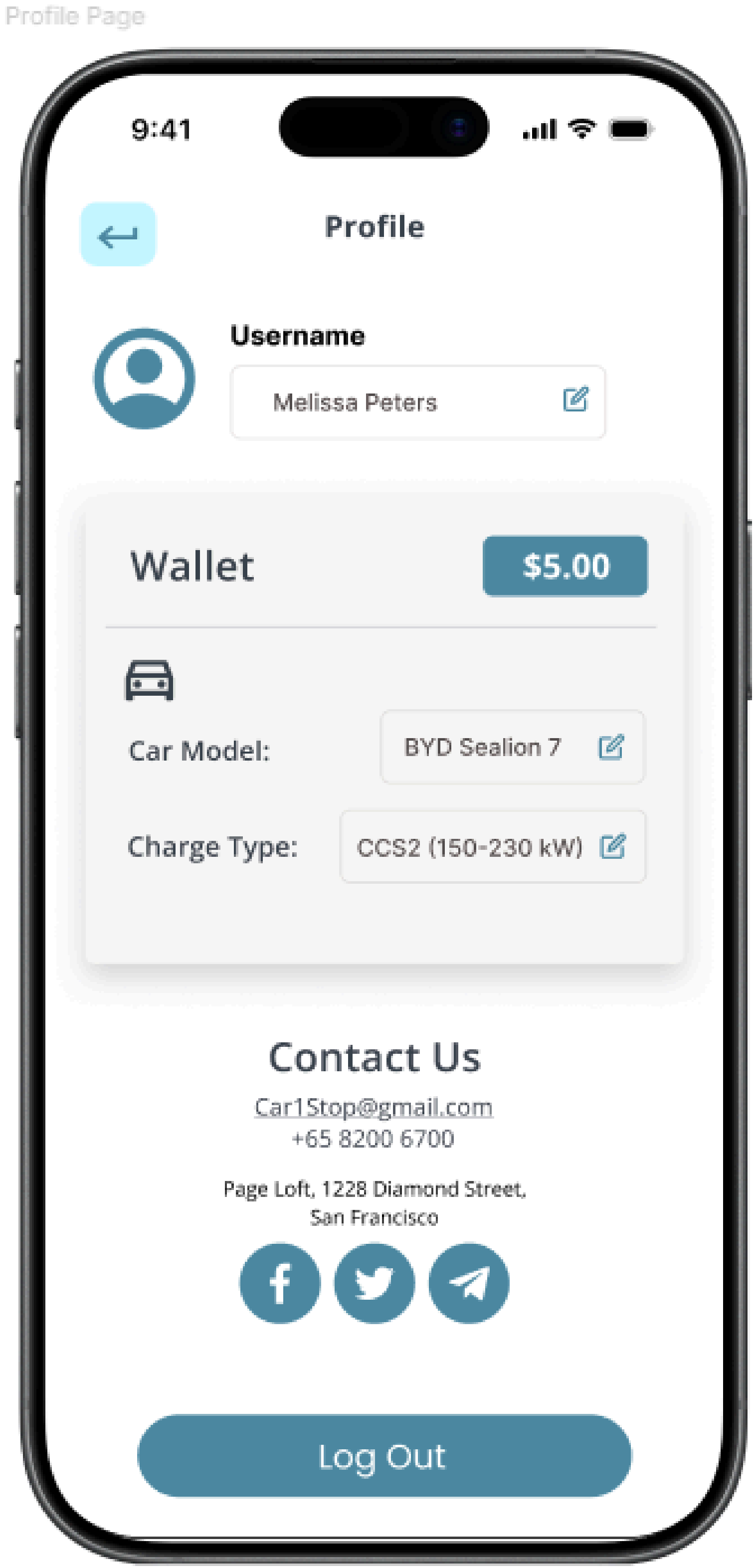
5.6 Top Up and Payment



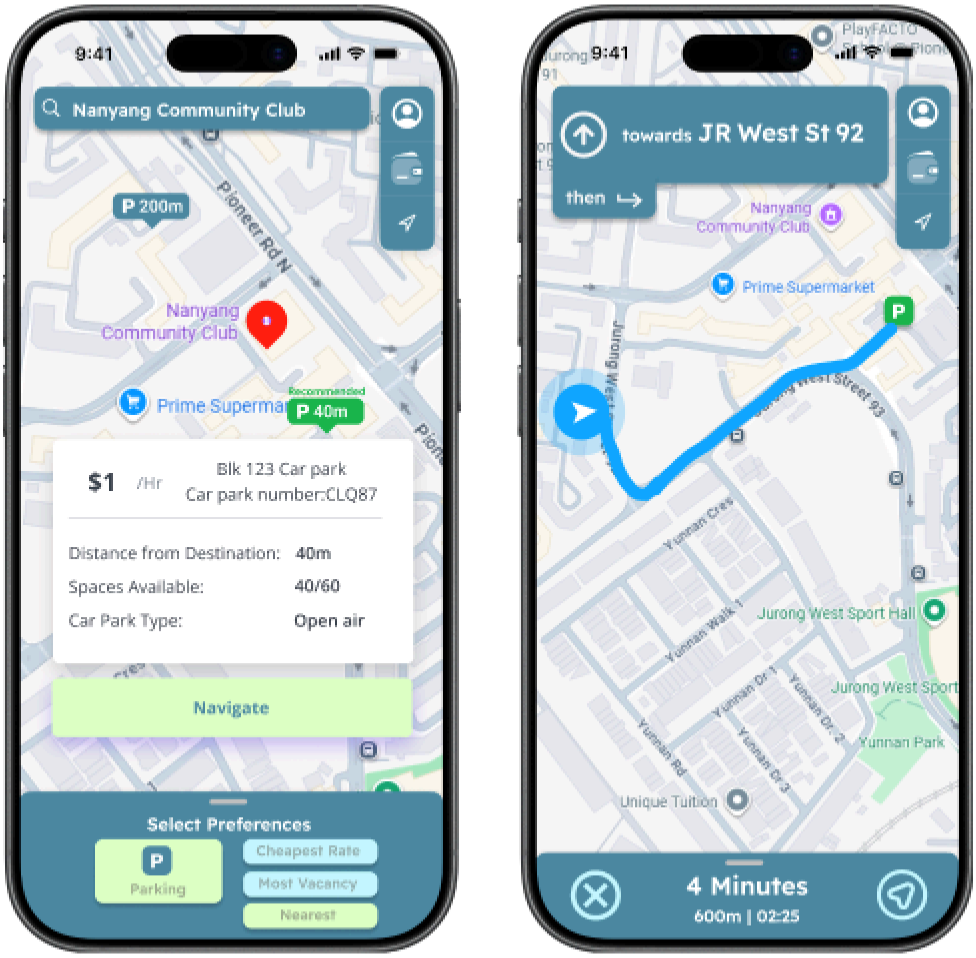
5.7 Booking and Top Up History



5.8 Profile Page



5.9 Routing and Route Cancellation



6. AI Critique

6.1 Original Prompt

As attached is my use case diagram and functional and non functional requirement. Critique the attached use case diagram image, with respect to how well it matches the attached requirements below.

6.2 AI Response

Nice draft — you've already captured most **core user goals** (search, recommend, book, top-up, navigate, profile).
But against your detailed requirement list, this diagram is currently a **partial match**.

My quick verdict: **about 55–65% traceability** to functional requirements.

What your diagram matches well

These are the strongest parts:

- **Core actors are present**: Registered User, Guest, External APIs, Payment Gateway, EmailService.
- **Main domain flows exist**:
  - Search destination / use current location / map view
  - Nearest/cheapest finding
  - Preference selection + option evaluation + ranking/recommendation
  - EV booking flow with wallet payment
  - Top up wallet
  - Start/stop navigation
  - Profile view/update + logout
- You correctly modeled some conditional logic, e.g. insufficient balance concept is present.

Where it does not match requirements well (by section)

1) User Account Management (Req 1.x) — partial

Present:

- Register, Login, Logout, Reset Password, View/Update Profile

Missing / weak traceability:

- **1.1 field-level registration rules** (email/username/password/car type+model) not shown

- **1.1.2 password policy** not shown
- **1.1.3 email format validation** not shown
- **1.2 email verification before activation + 24h link** not shown
- **1.3.3 authorized session creation** not explicit
- **1.3.4 descriptive login validation errors** not shown
- **1.5 forgot-password form/verify email exists/error** only partially represented
- **1.6 reset token invalidation + success redirect** not explicit
- **1.7.8 company contact details in profile** not shown

---

2) Location Search and Map Display (Req 2.x) — partial

Present:

- Request location permission, get current location, search destination, view map, determine search area

Missing / weak:

- **2.1.2 map zoom/pan/marker selection** not explicit
- **2.2.2 denied permission message for 3s** not shown
- **2.2.3 auto logout on denied permission** not shown
- **2.2.4 re-prompt on next app open** not shown
- **2.3.1 partial match by name/address** not shown
- **2.3.2 exact postal-code match** not shown
- **2.3.4 no matching destination message** not shown
- **2.3.6 center map to selected destination** only implied
- **2.4 strict 2km radius** only loosely implied by “Determine Search Area”

---

3) Preference-based Recommendation (Req 3.x) — mostly good

Present:

- Select main preference
- Evaluate options based on price/distance-time/availability
- Rank + highlight recommendations

Mismatch:

- Requirement says **top 3**; diagram says **“top 2-3”** (not exact match).
- “Recommended” tag behavior is not explicit as an output rule.

---

4) Carpark features (Req 4.x) — partial

Present:

- Find nearest/cheapest carpark
- Carpark pricing
- Carpark availability (colour indicators)

Missing / weak:

- **4.1 explicit filter carparks-only** not clear
- **4.3 click specific carpark -> detail view** partially implied, not explicit as one clear use case
- **4.4.3 sorted results list** implied via ranking, but not explicit carpark list
- **4.4.4 no carparks found message** missing

---

5) EV charging features (Req 5.x) — partial

Present:

- EV availability
- Book EV slot
- Select charger + time slot
- Cancel booking
- Insufficient balance check + block booking

Major missing:

- **5.3.1 EV detail fields** (adapter/provider/price) not explicit
- **5.5.1 booking confirmation notification** missing
- **5.5.2 fixed $5 booking fee** not explicit
- **5.5.2–5.5.6 grace period/no-show/release charger/realtime update/notifications** largely missing
- **5.6.1 prompt top-up when insufficient funds** not explicit
- **5.7 cancel only 30 min before reserved time** constraint not shown

---

6) Routing and Navigation (Req 6.x) — good

Present:

- Generate route
- Start navigation with updates
- Stop navigation

This section is one of your strongest matches.

---

7) Wallet & Payment (Req 7.x) — partial

Present:

- View wallet balance
- Top up wallet
- Payment gateway actor
- View booking transactions

Major missing:

- **7.2.1 exactly four predefined top-up amounts** not shown
- **7.2.2 select one amount at a time** not shown
- **7.2.4 exactly Card + PayNow methods** not shown
- **7.2.5/7.2.6 unique PayNow QR tied to amount** not shown
- **7.2.7 card fields (number/CVV/expiry)** not shown
- **7.2.8 Pay button validation logic** not shown
- **7.3 success flow (message + DB wallet update)** not explicit
- **7.4 error flow** missing
- **7.5 top-up + booking transaction history, latest 20, mandatory fields, no-history message** not explicit (diagram only shows booking transactions)

UML/use-case modeling quality issues in current diagram

These will likely be marked down in software engineering classes:

1. **Too cluttered / low readability**
   Heavy line-crossing makes traceability hard.
2. **Possible misuse of `<<include>>` vs `<<extend>>`**

- Some flows look reversed or semantically odd (e.g., “View Booking Transactions” extending payment flow).
- Rule of thumb:
  - `include` = mandatory reused subflow
  - `extend` = optional/conditional behavior

3. **Mixed abstraction levels**
   - Some are high-level goals (“Book EV charging slot”)
   - Some are low-level UI actions (“View wallet balance”, button-like behavior) This inconsistency hurts clarity.
4. **System boundary inconsistency**
   - “Determine Search Area” appears outside boundary.
   - Internal system behavior should sit within boundary.
5. **Non-functional requirements**
   - Use case diagrams generally do **not** capture NFRs (performance, availability, security constraints).
   - If your assignment expects NFR traceability, add supplementary spec tables/notes, not in main use-case ovals.

High-priority fixes (what to do next)

1. **Split into 3 diagrams** (cleaner + mark-friendly):
   - Account & Security
   - Search/Recommendation/Navigation
   - EV Booking & Wallet/Payments
2. **Add missing high-impact use cases**:
   - Verify Email Link (24h), Activate Account

- Validate Registration Fields / Password Policy / Email Format
- Handle Permission Denied (show 3s message + logout)
- Show No Results (search/carpark/EV)
- Booking Grace-Period Monitor + No-show Handling + Release Charger + Notify User
- Top-up via Card / Top-up via PayNow QR

3. **Make constraints explicit as notes** (instead of tiny ovals for every validation):
    - "Top 3 only"
    - "$5 booking fee"
    - "2 km radius"
    - "20 most recent transactions"
    - "30-minute grace period"
4. **Fix relationship directions (`include`/`extend`)** and keep only necessary ones.

—---

6.3 AI critique Justification

- **Quote: "Handling denied permission message for 3s not shown", "no matching destination message not shown "**
    - AI feedback highlights missing key important features such as handingly denied location permissions and also providing succinct error messages when no matches are found
- **Quote: "2.3.1 partial match by name/address not shown", "2.4 strict 2km radius only loosely implied"**
    - However, it could be more specific in recommending how to implement these features especially around partial matching and also regarding search radius adjustments.
- **Quote: "High-priority fixes (what to do next)"**
    - Overall, the feedback was extremely useful and we would be making the necessary changes. But it would be even more beneficial if it gave us actionable details and also pointers for user flexibility.